

1.0 Introduction

This project report details a plan for creating a Precision Agriculture Drone Solution designed specifically for small to medium-sized Irish Farms.

The nature of this project involves developing and deploying a crop monitoring system that uses drones fitted with high-resolution cameras and multispectral sensors. It aims to give farmers access to advanced technology without requiring significant initial investment.

Ireland is particularly well-suited for this project due to its strong agricultural sector, widespread use of small to mid-sized family farms, and growing national and EU-level emphasis on sustainability and smart farming technologies. The country's unique climate and landscape also make real-time, remote monitoring especially valuable for efficient resource management.

The project's primary purpose is to improve crop management practices on these farms. It seeks to achieve this by increasing crop yields while reducing water waste and other operational expenses. The project also aims to tackle farm labour shortages by providing an accessible technological alternative. Furthermore, the drone system is intended to facilitate early detection of pests and diseases, enable detailed monitoring of plant health, and optimise the application of resources like water and fertilisers. A key goal is to make this advanced farming technology affordable and readily available to the target farmers. It aims to decrease the reliance on time-consuming manual tasks like field inspections and cut costs associated with manual labour and delayed responses to crop issues. Ultimately, the project endeavours to improve overall farm efficiency through data-driven decision-making and promote sustainable agricultural methods by minimising the overuse of pesticides and fertilisers.

2.0 Integration Management

2.1 Project charter

1. Project title and description:

Precision Agriculture Drone Solution for Small to Medium-Sized Irish Farms:

Our project aims to provide a crop monitoring system using drones. We strive to target small and medium-sized farms. These drones will improve crop management, increasing yields by reducing water waste and crop-related costs. We try to address some of the issues farmers face, such as labour shortages, by providing them access to cutting-edge technology without high upfront costs. Using the drones equipped with high-resolution cameras and multispectral sensors, farmers can monitor plant health and detect pests and diseases early. At the same time, it's manageable and optimises the use of resources such as fertilisers and water.

2. Project Objectives

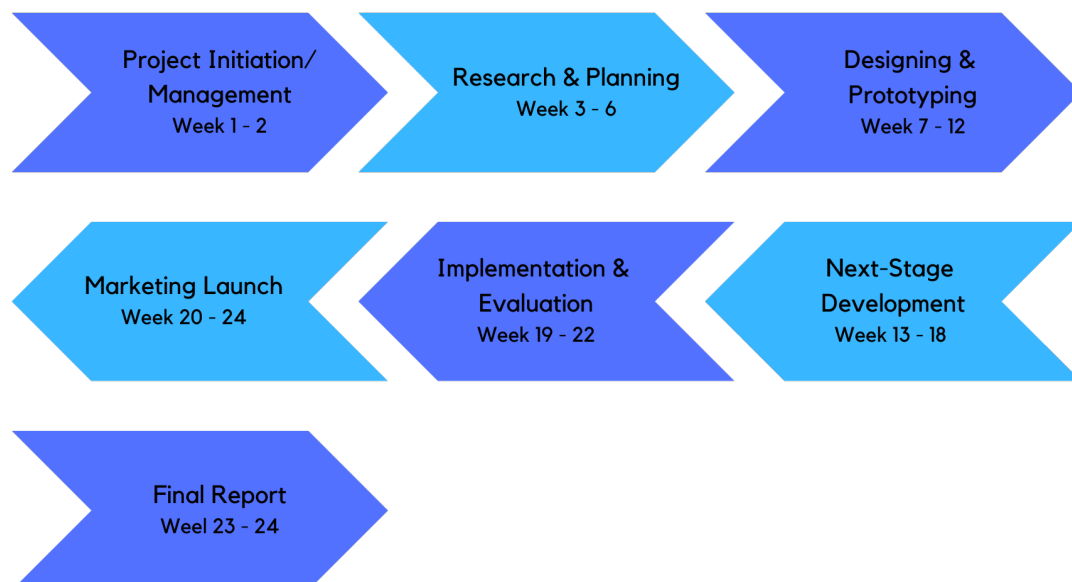
- Providing an affordable and accessible option for small to medium-sized farm owners.
- Reducing the amount of manual labour and time-consuming tasks (e.g., inspections) for farmers.
- Cutting manual labour costs, detecting early signs of issues like diseases and pests, and reducing the overall cost of late solutions.
- Improving crop yields and efficiency through data-driven decision-making.
- Promote sustainable farming by minimising pesticide and fertiliser overuse.

3. Project Success Criteria

- Stakeholders confirm satisfaction with performance, sustainability impact, and usability.
- The drone passes all regulatory testing and safety protocols (e.g., IAA and GDPR).
- At least 8 out of 10 successful flights and data captures in diverse Irish weather conditions.

- Over 80% of farmers report improved crop health monitoring and decision-making.
- At least 100 farmer sign-ups within 3 months of launch
- Subscription platform functions with a <5% data sync failure rate.
- Over 80% of training participants report confidence in drone operations.
- Efficient data sync between the data platform and farmers with a downtime of <5%.

4. Summary Schedule



5. Stakeholders: Roles and Responsibilities

External Stakeholders:

- Farmers: End-users benefit from improved crop management.
- Investors & Funders: Provide potential financial support.
- Regulatory Authorities: Ensure legal compliance in drone operations.
- Technology Providers: Supply sensors, software, and drone components.
- Agricultural Consultants: Guiding the integration of drone data with suitable farming practices.

- Agricultural cooperatives and associations: Promoting collaboration in precision farming research and adoption.
- Government & Policy Makers: Ensure alignment with EU sustainability goals.
- Insurance Companies: Offer coverage for the drone and its tasks.

Internal Stakeholders:

- Project Manager: Oversees timeline, budget, quality, and team performance.
- Project Team Members: Individuals responsible for various tasks, including the design, software and testing.
- Software Developers: Build data collection and analysis software.
- Drone Engineers & Developers: Designing, developing and testing drones.
- Data Analysts: Interpret collected data to make it relevant to farmers.
- Sales & Marketing Agents: Promote drone solutions to target users.
- Finance & Accounting Team: Manage project spending and forecasts.
- Legal Team: Ensuring the project complies with laws, regulations and policies.
- Customer Support & Training Team: Assist farmers and ensure the technology is used effectively.

6. Constraints and Assumptions

Constraints:

- The project must comply with Irish and EU regulations for drone use.
- Budget limitations for small farmers must remain affordable.
- Drone performance may be weather-dependent (rain, wind, ...).
- Battery life and drone range may limit large-scale use.
- Direct integration with existing farm machinery.
- Variable rural internet connectivity for data uploads.

Assumptions:

- Farmers are ready to adopt new technologies if the benefits are clear.
- Drone-collected data is reliable and leads to actionable insights.
- Insurance and maintenance are manageable by farmers.

2.2 SWOT Analysis

Strength	Weaknesses
• Expert drone and software development team • High-resolution cameras and multispectral sensors for precise monitoring • Ability to optimise water, fertiliser, and pesticide use, leading to cost savings • Alignment with EU sustainability goals and regulations • Scalable monitoring system that reduces manual labour • Potential for strong market differentiation in precision agriculture • Data-driven insights improving farm productivity and efficiency	• High initial investment required for drone acquisition and maintenance • Dependence on favourable weather conditions for accurate data collection • Farmers may require technical training and adaptation period • Ongoing software subscription and cloud storage costs • Potential issues with drone maintenance, wear, and damage
Opportunities	Threats
• Rising demand for sustainable and precision agriculture solutions • Government and EU funding for sustainable farming initiatives • Potential market expansion due to labour shortages in the agricultural sector • Increasing interest from younger farmers in adopting new technology	• Data security and privacy concerns regarding farm data collection • Regulatory challenges and compliance with evolving drone laws • Rapidly changing technology landscape requiring continuous innovation • Market competition from other precision agriculture technologies

• Potential collaborations with agricultural cooperatives and research institutions • Growing demand for AI-based data analysis in agriculture • Expansion into insurance and financial services for drone usage • Ability to develop new business models such as leasing or pay-per-use drone services	• Environmental risks such as extreme weather affecting drone functionality • Resistance from traditional farmers hesitant to adopt new technology • Potential risks of drone interference with birds, aircraft, or other obstacles
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3.0 Scope Management

Scope Description:

The project aims to develop and implement a high-tech drone solution tailored to the needs of small and medium-sized farmers in Ireland. The drone will assist in crop health monitoring, data-driven decision-making, and sustainable resource management. The solution includes but isn't restricted to the drone, a user training and licensing package, a subscription-based analytics and storage service, and a cloud-based data storage and analysis backend. This project assumes a willingness among Irish farmers to adapt to new technologies.

Deliverables:

- Fully functional agricultural drone with integrated multispectral and HD cameras
- Farmer training and licensing materials
- Maintenance manual and repair guidelines
- User manuals and technician training materials
- Analytics subscription service with a web-based analytics dashboard
- Cloud storage infrastructure with backup and redundancy
- Real-time data processing and sync pipeline
- API documentation for frontend-backend communication
- Data privacy and security protocol documentation (GDPR compliant)
- Encrypted data management system
- Drone control mobile/web interface
- Image processing software for plant health and pest analysis
- Marketing launch campaign
- Customer support portal (chatbot, help centre, ticketing system)

Acceptance Criteria:

- The drone passes all regulatory testing and safety protocols (e.g., IAA and GDPR).
- At least 8 out of 10 successful flights and data captures in diverse Irish weather conditions.
- Over 80% of farmers report improved crop health monitoring and decision-making.

- At least 100 farmer sign-ups within 3 months of launch.
- Subscription platform functions with a <5% data sync failure rate.
- Over 80% of training participants report confidence in drone operations.
- Efficient data sync between the data platform and farmers with a downtime of <5%.

Exclusions:

- Cost of initial integration: Any technical or operational costs associated with integrating the drone system with a farmer's existing equipment or digital tools (e.g., ERP systems, irrigation controls) are not covered by this project.
- Direct integration with existing farm machinery: Custom software or hardware adaptations for compatibility with specific farm machinery brands or types (e.g., tractors, sprayers) are not included.
- Custom drone modifications for niche farm types: The drone model is designed for general agricultural use and does not include tailored adaptations for highly specialised farms (e.g., vineyards, orchards with steep terrain).
- Complete automation of drone operations (beyond pilot-assisted flight): The system assumes a pilot-assisted workflow. A full autonomous operation is outside the current scope.
- Limited project budget and timeline: Custom requests beyond the defined deliverables may not be feasible within the current constraints.
- Regulatory boundaries around drone flight operations: The system is designed to comply with Irish and EU drone regulations. Features that would require special exemptions are excluded.
- Variable rural internet connectivity for data uploads: While basic offline functionality is included, solutions to poor connectivity (e.g., satellite uplinks or rural broadband installation) are not part of this project.
- Weather limitations for flight testing: Field testing is limited to conditions considered safe and practical. Adverse weather may delay or restrict test deployments.

3.1 Work Breakdown Structure (WBS)

Agricultural Drone Deployment Project

1. Project Management
 1. Planning
 2. Meetings
 3. Administration
2. Drone Development
 1. Hardware Design & Prototyping
 2. Testing & Iteration
 3. Manufacturing Preparation
 4. Maintenance Plan
3. Software Development
 1. Drone Operation Interface
 2. Image Processing & Analysis
 3. User Interface
4. Data & Backend Infrastructure
 1. Cloud Data Storage
 2. Data Sync & Processing
 3. Security & Compliance
5. Training & Licensing
 1. Training Content Development
 2. Certification & Licensing
 3. Onboarding Support
6. Analytics Subscription Platform
 1. Dashboard & Metrics
 2. Account & Billing System
 3. User Support System
7. Marketing & Outreach
 1. Branding & Materials
 2. Target Campaigns
 3. Events & Partnerships

8. Regulatory Compliance

1. Aviation Compliance
2. Environmental Standards
3. Data Privacy & Legal

3.2 WBS Dictionary

WBS Item Number	WBS Item Name	Description
1.	Project Management	Planning, resourcing, scheduling, stakeholder communication, and risk management across all phases.
1.1	Planning	Develop the overall project plan, schedule, budgeting, and resource allocation.
1.2	Meetings	Organise and document team meetings, stakeholder presentations, sprint planning, and review sessions to ensure collaboration and communication.
1.3	Administration	Manage documentation, reporting, procurement coordination, compliance tracking, timesheets, and communication logs.
2.	Drone Development	Design, prototype, test, and prepare for manufacturing of the drone hardware.
2.1	Hardware Design & Prototyping	Design and develop drone hardware including chassis, sensors, battery systems, and weather-proofing features.
2.2	Testing & Iteration	Conduct lab and field tests for performance evaluation. Refine hardware based on test results and farmer feedback.
2.3	Manufacturing Preparation	Finalise the Bill of Materials, identify suppliers, establish assembly protocols, and conduct quality assurance planning.
2.4	Maintenance Plan	Develop guidelines for regular maintenance, diagnostics, and repairs. Includes user manuals and technician training materials.

3.	Software Development	Build web applications for drone operation and farmer interfaces.
3.1	Drone Operation Interface	Build a user-friendly web interface to control the drone, manage flight paths, and monitor real-time performance.
3.2	Image Processing & Analysis	Implement algorithms for plant health monitoring, pest/disease detection, and terrain mapping using drone-collected data.
3.3	User Interface	Design the front-end of the software platform, emphasising usability, simplicity, and intuitive dashboards for non-technical users.
4.	Data & Backend Infrastructure	Setup of cloud systems, backend data pipelines, security protocols.
4.1	Cloud Data Storage	Set up scalable and secure cloud storage to manage image, sensor, and user data. Ensure redundancy and quick access.
4.2	Data Sync & Processing	Build automated pipelines to process drone data and sync it with user accounts in near real-time.
4.3	Security & Compliance	Implement security protocols including encryption, GDPR-compliant user consent, and secure authentication systems.
5.	Training & Licensing	Design and deploy training materials, user guides, and certification workflows.
5.1	Training Content Development	Create instructional content covering drone usage, basic troubleshooting, and software features. Includes video and written materials.
5.2	Certification & Licensing	Guide farmers through Irish Aviation Authority (IAA) certification processes and ensure compliance with flight regulations.
5.3	Onboarding Support	Provide live training sessions, Q&A helpdesks, and initial operational assistance during first-time drone use.

6.	Analytics Subscription Platform	Create a SaaS model with a data dashboard, billing, and support tools.
6.1	Dashboard & Metrics	Develop analytics dashboards that offer insights into crop health trends, irrigation efficiency, and disease predictions.
6.2	Account & Billing System	Integrate payment systems for managing subscriptions and user accounts. Include billing tiers and free trial features.
6.3	User Support System	Provide support tools such as help center articles, chatbot assistance, and direct support ticketing systems.
7.	Marketing & Outreach	Branding, advertising, demo events, partnerships, and farmer engagement.
7.1	Branding & Materials	Design brand identity, promotional materials (videos, brochures), and a website to introduce the product to the market.
7.2	Target Campaigns	Launch marketing efforts focusing on agricultural co-ops, rural communities, and government programs for small farmers.
7.3	Events & Partnerships	Participate in agricultural expos, host demo events, and build partnerships with industry groups and tech providers.
8.	Regulatory Compliance	Ensure project adheres to legal, aviation, environmental, and data privacy laws.
8.1	Aviation Compliance	Ensure the drone system meets all operational, safety, and registration standards set by the IAA and EU airspace laws.
8.2	Environmental Standards	Design drone usage protocols that align with EU CSRD and national sustainability goals, including emissions tracking.
8.3	Data Privacy & Legal	Draft user agreements, privacy policies, and legal disclaimers to ensure full compliance with GDPR and consumer protection laws.

4.0 Stakeholder Management

Stakeholders

External Stakeholders:

- Farmers: Mostly family businesses, from small to medium-sized
- Sponsors: Companies and organisations with a big urge for innovation and sustainability in the agriculture sector
- Competitors: Analytic companies, agriculture machinery companies, agriculture consulting companies
- Locals: People living in the area
- Regulatory Authorities
- Technology Providers/ Suppliers
- Agricultural Consultants
- Government
- Insurance Companies

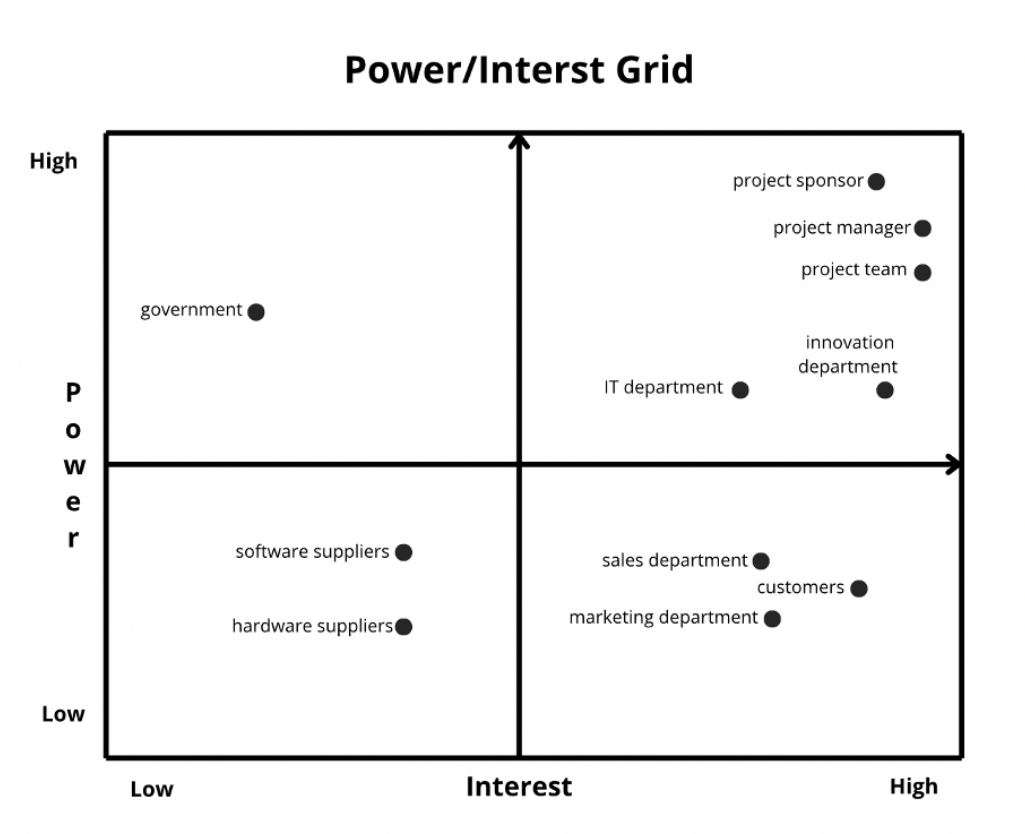
Internal Stakeholders :

- Project Manager
- Project Team Members
- Software Developers
- Drone Engineers & Developers
- Data Analysts
- Sales & Marketing Agents
- Finance & Accounting Team
- Legal Team
- Customer Support & Training Team
- Innovation Department

4.1 Stakeholder register

No.	Name	Position	Internal/ external	Project Role	Contact Information
1	George	IT architect	Internal	Team member	george@agrodrone.ie
2	Abby	CFO	Internal	Board member	abby@agrodrone.ie
3	Johnatan	CIO	Internal	Senior manager	johnatan@agrodrone.ie
4	Bob	VP of Delivery	Internal	Project sponsor	bob@agrodrone.ie
5	Thomas	Architect	Internal	Innovation department member	thomas@agrodrone.ie
6	Alan	Customer	External	Questionnaire donor	alan@farmer.ie
7	Peter	Legal inspector	External	Government representative	peter@government.ie
8	Christian	Materials supplier	External	Drone hardware supplier	christian@supplier.ie
9	Brad	Data analyst/ERP software vendor	External	Software supplier	brad@vendor.ie
10	Kevin	External Agro consultant	External	Project manager	kevin@agroconsulting.ie

4.2 Power / Interest Grid



4.3 Expectation Management Matrix

Measure of Success	Priority	Expectations	Guidelines
Scope	2	The scope defines mandatory requirements but also each and every implementation for every farmer can be different on some aspects.	Focus on the base guidelines and its conventions to keep it as cost efficient and simple as possible but make enough individual changes to customers for personalizing.
Time	3	The quality, cost and customer satisfaction are much more important as the only real factor that urges the project is to not fall behind competitors.	The project sponsor and board members must always be in perspective about the status of the project to be able to

			communicate it to stakeholders.
Cost	2	The project is a big part of the company both financially and portfolio wise. The cost of execution is important, but funds can be made available, not the most important aspect.	Cost changes from the original IT development and hardware development financial plan should always be brought to the senior manager to discussion.
Quality	1	The most important aspect of the project, since both the drones and their software's (with the analysing features) are crucial to satisfy our customers and to create real value.	The drones both software and hardware should be meeting the agreed standards. Quality is the one thing where we do not compromise.
Customer Satisfaction	1	The main aspect why quality is so important is to satisfy our customers, this is the goal of the project and to create actual value to them to transform this project into a stable product/service in the future.	The customers experience and satisfaction is prior. It is important that we help them build their system with our best knowledge. If they don't know how to scale and make the best out of our system, our duty is to consult with them and personalize the system for their needs.
Technology	1	Technology is important to create the most cost efficient and accurate solution for our customers to bring out the best of their resources.	The innovation team should always communicate with the IT and quality management team. These departments should be in harmony and agreement at all times as their

			work together creates our product/service and makes it unique and market leading.
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5.0 Communication Management

Summary

This Communications Plan includes the communication goals, stakeholder needs, delivery channels, and a feedback system. The strategy aims to keep everyone updated and involved through our Drone project.

Background

This communications plan is developed to manage all information while implementing a drone-powered crop monitoring system. The technology will assist small—to medium-sized farmers in improving crop management, operating expenses, and time efficiency.

Communication Goals

The main goal is to inform stakeholders of project risks, progress, and results. Farmers should be kept informed of the drone's performance and benefits. Regulators should be in open contact to guarantee safety and legal requirements.

Stakeholders

The project's stakeholders include both internal and external groups. External stakeholders include farmers, investors, regulators, technology suppliers, and agricultural consultants. Internal stakeholders are project managers, team members, software developers, drone engineers, and marketers.

Stakeholder Requirements

Role	Information	Timeframe
Farmers	Updates on system use, crop health data, training, support	Weekly, Monthly, Quarterly
Investors	Financial performance, project milestones.	Monthly
Regulatory Authorities	Compliance updates, drone operation reports, legal documentation	1 st each month

Technology Providers	Updates on hardware and software requirements, system specifications	As needed
Agricultural Consultants	Feedback on system integration with farming practices	Monthly

Key Messages

The project's main messaging highlights the drone system's benefits, development, and environmental sustainability. Each stakeholder group will receive customised core messages.

Farmers: Our drone system allows you to monitor crop health efficiently, detect issues early, and optimise water and fertiliser use, reducing costs. Investors: "This project offers a scalable, innovative solution to some of the biggest challenges in modern farming, with strong growth potential and a significant positive impact on sustainability. Regulatory Authorities: Our system follows all regulations and standards, promoting sustainable agricultural practices aligned with European environmental goals.

Channels

Method	Frequency	Goal	Audience
Emails	Weekly	Share project status, discuss potential delays or issue	Project Team, Investors, Regulatory Authorities
Internal meetings	Weekly	Discuss project progress, assign tasks, address blockers	Project team
Project Review	At milestones	Present deliverables, gather feedback, and discuss next steps	Project Team, Investors
Newsletters	Monthly	Provide system updates, performance data, milestones and wins	Farmers, Investors, internals.
Training Workshops	Quarterly	Teach farmers how to use the drones, optimize their use	Farmers

Communications Events

Event	Description	Purpose
Project Team Meeting	Meeting involving all team members, to discuss the work in-progress / recently completed / coming up	To keep the team informed of the project status and ensure that issues, risks or changes are raised early on.
Project Board Meeting	Formal meetings held with the Project Board to assess the overall status of the project.	To determine whether the project has been completed and met the final requirements of the customer.
Phase Review Meeting	Formal meeting held at the end of each phase, to determine whether the quality of the deliverables produced is satisfactory.	To control the progress of the project through each phase in the lifecycle and boost its chance of success.
Change Control Meeting	A meeting dedicated to assessing and approving or rejecting proposed changes to the project scope, budget, or timeline.	To evaluate proposed changes, understand their impact on the project, and make decisions on how to proceed.
Customer Feedback Session	A meeting or series of discussions with end users or customers to gather feedback on the project deliverables, usability, and satisfaction.	To ensure the final deliverables meet the customer's expectations and gather valuable insights for improvements or refinements.
Cross-Departmental Coordination Meeting	A meeting where representatives from various departments come together to discuss project alignment and cooperation.	To ensure that departments are aligned, fostering collaboration across teams and minimizing miscommunication or delays in cross-functional work.

Feedback Measures

We will use various methods to gather feedback. These feedback measures will help us understand how well the communication event was received and whether it met the audience's needs. For instance, we will send short surveys after key meetings or events. These will ask questions about how clear, timely, and helpful the information was. This approach allows us to get a quick sense of the effectiveness of the communication.

In addition to the surveys, feedback forms will be provided to stakeholders. These forms will allow them to rate the communication based on clarity, relevance, and how well the information was presented. If there are any issues or concerns with the communication, stakeholders can submit complaints through a formal complaints form. This ensures that any dissatisfaction is formally captured and addressed.

Success Criteria

Once feedback is collected, we will compare it to a set of success criteria to evaluate whether the communication strategy was successful. We will first assess if the message reached the target audience, ensuring that the right people received the information. The next step is to confirm that the communication was sent through the planned channels. The timing of the communication is also crucial. We will measure whether the message was delivered on schedule and whether it was effective in achieving its original goals. Another factor we will consider is whether the message was clear, properly formatted, and engaging for the audience. Additionally, we will look at whether the message achieved the desired effect.

It is equally important that the message is perceived as honest and trustworthy. We will assess whether stakeholders felt that the communication was transparent. Lastly, we will examine whether there were any complaints.

6.0 Schedule Management

Our project will follow a structured schedule to ensure timely completion while maintaining high-quality deliverables.

6.1 Key Phases

1. Project Initiation/ Management

- Defining project scope, objectives and deliverables.
- Identifying stakeholders and project roles.
- Conducting feasibility studies and risk assessments.

Period: week 1 – week 2

2. Research & Planning

- Researching drone technologies, clearing device integrations, and identifying development requirements.
- Assessing the regulatory requirements for drone usage in our specific sphere.
- Developing a detailed WBS.

Period: week 3 – week 6

3. Design & Prototype

- Developing an initial drone prototype.
- Creating the first version of software for data processing.
- Conducting pilot tests to measure performance.

Period: week 7 – week 12

4. Next-Stage Development

- Improving drone capabilities and customising them for agricultural uses.
- Improving software functionality and visualisation.
- Creating data and backed infrastructure
- Designing training content and licensing material
- Conducting pilot tests in chosen farms.

- Building a subscription dashboard and a billing system.

Period: week 13 – week 18

5. Implementation & Evaluation

- Ensuring regulatory compliance.
- Deploying drones for real-world usage on sample medium-sized and small farms.
- Gathering farmer feedback and optimising performance.
- Evaluating environmental and cost-saving impact.
- Training farmers and guiding them through IAA licensing.
- Providing onboard support.

Period: week 19 – week 22

6. Marketing Launch

- Branding.
- Demo events.
- Campaign execution.

Period: week 20 – week 24

7. Final Report

- Document project results, challenges and solutions.
- Presenting performance analysis and recommendations.
- Conducting final stakeholder review and closure.

Period: week 23 – week 24 (Parallel)

Task	Responsible Role
Project Initiation	Project Leaders and Sponsors
Research & Planning	Project Manager & Analysts
Design & Prototype	Drone Engineers and Software Developers
Next-Stage Development	Field Technicians and Analysts

Implementation & Evaluation	Farmers and Consultants & Project Team
Final Report	Everyone

Risk Mitigation & Contingencies

- **Regulatory Delays**

Early engagement with authorities is required.

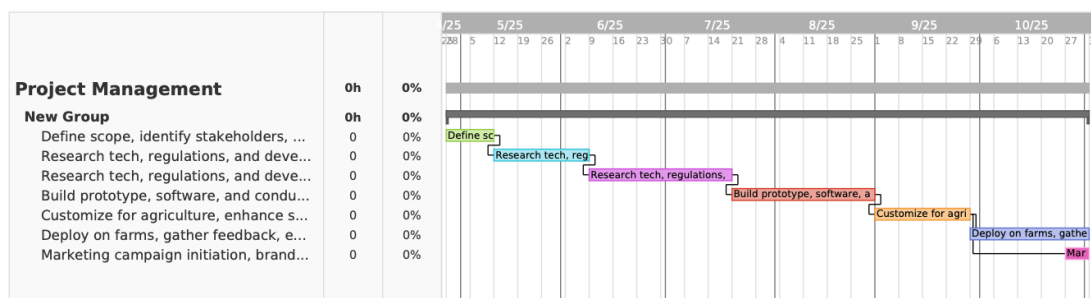
- **Technical Failures**

Multiple testing phases are required.

- **Data Inaccuracy**

Regular calibration of software adjustments is required.

6.2 Grantt Chart



7.0 Risk Management

Introduction

Managing risks is essential for our agricultural drone project. This section outlines how we'll identify, analyse, and handle potential issues that could affect our project goals. We've created a risk register to track concerns, their likelihood of happening, their potential impact, and what we plan to do about them.

Risk Identification and Analysis

Our risk register below shows potential problems across various categories - from regulatory to technical difficulties. We've rated each risk's likelihood and impact on a 1-5 scale (1 being minimal, 5 being severe) and calculated a risk score by multiplying these values to help us prioritise.

7.1 Risk Register

Risk ID	Category	Description	Project Connection	Likelihood (1-5)	Impact (1-5)	Score	Owner
R01	Regulatory	Changes to Irish/EU drone laws requiring us to adjust operations	Connected to regulatory stakeholders and compliance needs from our charter	3	4	12	Legal Team
R02	Technical	Our technology becoming outdated as drone tech advances	Links to technology providers and the changing tech landscape noted in our SWOT	2	3	6	Drone Engineers

R03	Environmental	Bad weather disrupting flights and data collection	Noted in our SWOT and constraints section	4	3	12	Project Manager
R04	Security	Farm data security and GDPR compliance issues	Highlighted as a major threat in our SWOT	3	5	15	IT Security
R05	Market	Farmers rejecting or slowly adopting our technology	An assumption in our charter and threat in our SWOT	4	5	20	Marketing
R06	Financial	Running short on funding, affecting timeline and scope	Involves our investors and funders	2	4	8	Finance
R07	Operational	Supply problems affecting component availability	Relates to our technology providers	3	3	9	Procurement
R08	Technical	Software development delays or problems	Involves our software development team	3	4	12	Dev Lead
R09	Operational	Drones breaking or malfunctioning	Partially covered by our insurance stakeholders	3	3	9	Drone Engineers

		ng, causing data loss					
R10	Technical	Problems syncing data between drones and cloud	Affects our backend infrastructure work	3	4	12	Software Team
R11	Operational	Poor internet in rural areas affecting data transmission	Related to our rural connectivity constraint	4	4	16	Data Team
R12	Environmental	Drones disturbing local wildlife	Connected to our environmental standards work	2	3	6	Environmental Team

Response Planning

We've created specific plans for each risk, focusing extra attention on the high-scoring ones.

Our approaches include:

- Avoiding risks by changing our plans
- Reducing risks by making them less likely or less harmful
- Creating backup plans for if risks happen
- Shifting risks to third parties (like through insurance)
- Sometimes, accepting smaller risks

Our Biggest Concerns (Score ≥ 12)

1. Farmers Not Adopting Our Technology (R05, Score: 20)
 - a. What we'll do:
 - i. Create easy-to-follow training with hands-on sessions
 - ii. Run pilot programs with early adopters to show real results

- iii. Offer flexible payment options to reduce financial barriers
- iv. Make our interfaces simple and farmer-friendly
- v. Find respected farmers to test and promote our solution
- vi. Create a network of farmers who can share experiences

2. Rural Internet Problems (R11, Score: 16)

- a. What we'll do:
 - i. Map out connectivity in target areas before deployment
 - ii. Build offline features that work without constant internet
 - iii. Process critical data on the drone itself when possible
 - iv. Compress data for easier transmission
 - v. Create backup options for areas with terrible connectivity
 - vi. Talk to telecoms about improving rural coverage

3. Data Security Issues (R04, Score: 15)

- a. What we'll do:
 - i. Use strong encryption and security measures
 - ii. Follow GDPR rules completely and get certified
 - iii. Test our security regularly through outside experts
 - iv. Give farmers clear information about how their data is used
 - v. Make sure farmers maintain ownership of their data
 - vi. Create steps to follow if there's ever a data breach

Other Important Risks

4. Regulatory Changes (R01)

- a. What we'll do: Keep track of rule changes; design drones we can modify if needed; build relationships with regulatory officials; join industry groups to stay informed.

5. Weather Problems (R03)

- a. What we'll do: Schedule flights during good weather; make our drones more weather-resistant; develop backup methods like ground sensors; create smart scheduling that adapts to forecasts.

6. Software Development Delays (R08)

- a. What we'll do: Use agile methods to catch problems early; test thoroughly and review code regularly; hire experienced developers; have backup developers available if needed.
- 7. Data Sync Problems (R10)
 - a. What we'll do: Create reliable data transfer methods; allow offline storage when needed; build multiple ways to sync data; test in various connectivity scenarios.

Lower Priority Risks

- 8. Supply Chain Issues (R07, Score: 9)
 - a. What we'll do: Use multiple suppliers; identify backup suppliers; keep extra inventory of critical parts.
- 9. Drone Damage (R09, Score: 9)
 - a. What we'll do: Get good insurance coverage; create regular maintenance schedules; train operators well; keep backup drones ready.
- 10. Funding Problems (R06, Score: 8)
 - a. What we'll do: Secure funding from different sources; plan implementation in phases; watch our budget closely; keep some money in reserve.
- 11. Technology Outdated (R02, Score: 6) & Wildlife Disruption (R12, Score: 6)
 - a. What we'll do: Use modular designs we can upgrade; create wildlife-friendly flight paths; stay current on new technologies.

Monitoring and Control

We'll keep track of risks throughout the project using these methods:

- 1. Regular Risk Meetings: Every two weeks, we'll review current risks, look for new ones, and adjust our plans as needed.
- 2. Tracking Key Metrics: We'll set up specific measurements to tell us if our risk strategies are working.
- 3. Linking with Change Management: Whenever the project changes, we'll check if that creates new risks.

4. Keeping Stakeholders Informed: We'll talk openly with stakeholders about major risks and what we're doing about them.
5. Clear Escalation Steps: We'll define when and how to bring risk issues to higher management.
6. Learning from Experience: We'll document what happens with each risk to help with future projects.

8.0 Cost Management

8.1 Cost Estimation Approach

For this project, the bottom-up approach has been chosen to estimate costs. This method involves identifying each individual work package or deliverable in the Work Breakdown Structure (WBS), estimating the cost of each, and then aggregating them to form a comprehensive project cost estimate. This approach ensures accuracy because it considers the unique resource requirements, labour, and materials for each element of the WBS.

8.2 Cost Baseline

The Cost Baseline is the approved time-phased budget against which project performance can be monitored. It includes the planned expenditure across all work packages and incorporates contingency reserves for identified risks. This baseline will be used to manage cost performance throughout the project lifecycle.

8.3 WBS-Based Cost Estimate

WBS Element	Description	Estimated Cost (€)
1. Project Management	Stakeholder meetings, documentation	2,700
2. Drone Development	Hardware design, prototyping, testing, QA, and prep for manufacturing	10,000
3. Software Development	Web interface for drone control, image processing, and user dashboard design	9,000
4. Data & Backend Infrastructure	Cloud storage, data pipelines, real-time syncing, security protocols	6,500

5. Training & Licensing	Instructional content, certification processes, onboarding support	2,800
6. Analytics Subscription Platform	SaaS setup, billing system, analytics dashboards, user support tools	4,500
7. Marketing & Outreach	Branding, campaigns, events, and partnership development	3,500
8. Regulatory Compliance	Aviation, environmental, and data privacy/legal compliance	1,800
9. Contingency Reserve (10%)	Buffer for unforeseen issues	4,280
Total Estimated Cost		45,080

9.0 Quality, Resource and Procurement Management

9.1 Raci chart (R-responsible, A-accountable, C-consulted, I-informed)

	Project manager	Software developer	Engineer	Agro consultant	Sales-person	Project sponsor
Software and app	R	A	C	I	I	I
Drone hardware	R	C	A	I	I	C
Application builds up features	R	C	C	C	I	C
Monitoring customer needs	R	I	I	A	A	C

9.2 Procurement Management

Most of the skillset is available in the company, however, if needed, Software development partners can be hired for specific tasks. In this case, the company would mainly use nearshoring because it is important to consult about the specific requirements and features in person to achieve the best quality possible.

9.3 Contract Types

Our contracts with customers will be Time and Material contracts. This is necessary because one part of our product is fixed (this includes the drones and the software, which consists of a license price for the software and a unit price for the drones), while the other part involves the specifications that personalise the software for each and every farm. These specifications and software modifications incur an hourly cost, as our developers and consultants will be working on it, resulting in a significant amount of labour costs.

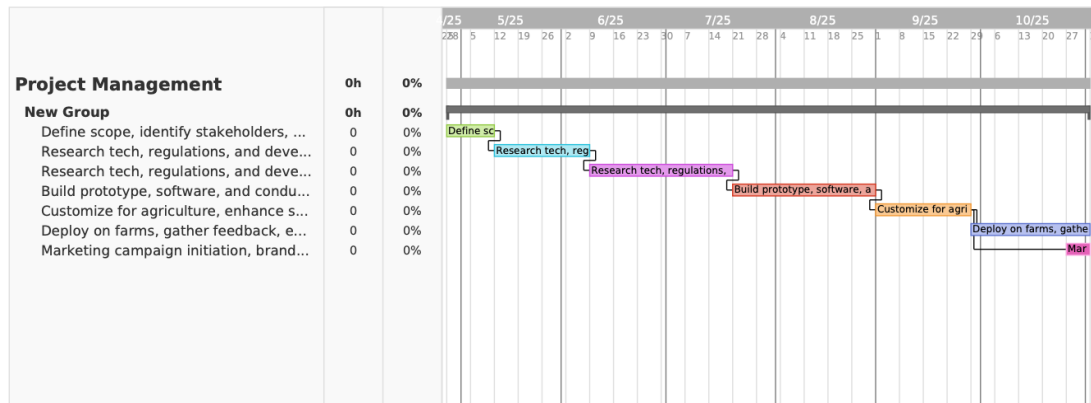
9.4 Human Resource Management

As our team will implement many systems driven by drones and this software, they must develop teamwork skills, knowledge about their job, and customers. They should not only be

able to fulfil their direct requests but also know what their customers are thinking about, even though sometimes these customers cannot explain themselves properly.

10.0 Software Tool Implementation

10.1 Gantt Chart



For effective project scheduling and task tracking, we chose TeamGantt, a dedicated project management software designed specifically for creating Gantt charts. Unlike basic productivity tools like word processors and spreadsheets, TeamGantt is built for planning and visualising project timelines, dependencies, and resource allocation, which aligns with our project's goals.

TeamGantt was applied to plan and visualise the entire project workflow from research to marketing. Here's how it was used:

1. Task Breakdown: The project was divided into clear phases:
 - a. Scope definition
 - b. Research
 - c. Prototype building
 - d. Customisation
 - e. Deployment
 - f. Marketing
2. Timeline and Dependencies: Each task was scheduled based on realistic durations, with dependencies set to reflect the natural order of the workflow.
3. Progress Tracking: A percentage of completion was added to monitor the progress in each phase.

4. **Hours Count:** An hour metric added to show the total amount of time expected to achieve the phase's goal. This can help in many aspects, including schedule management, resources, and financial allocations.

A more detailed example of a sample phase can be seen in the image below.

The screenshot displays a project management interface for a phase titled "Define scope, identify stakeholders, feasibility & risk assessment". The phase is 8% complete, with a start date of 4/28/25 and an end date of 5/9/25. The interface is divided into two main sections: a left sidebar and a right details panel.

Left Sidebar:

- People:** Includes links for "Assign people +" and "Assign labels +".
- Labels:** A section for assigning labels.
- Notes:** A section with a rich text editor and a list of notes:
 - Defining project scope, objectives and deliverables.
 - Identifying stakeholders and project roles.
 - Conducting feasibility studies and risk assessments.
- Comments:** A section with a text input field labeled "Add a comment or file..." and a "Send" icon.
- Files:** A section with a dashed box for uploading files, labeled "Upload a file" and "Click to browse, or drag & drop files here". It also includes a "From Dropbox" link.
- History:** A section for viewing the history of the phase.

Right Details Panel:

- Details:** A section with various fields:
 - Points:** A field for entering points.
 - Duration:** A field showing "10 days".
 - Parent Group:** A dropdown menu showing "New Group".
 - Type:** A dropdown menu showing "Task".
 - Color:** A dropdown menu showing "Golfers Green".
- Predecessors:** A section with an "Add" button and the text "No predecessors added yet".
- Successors:** A section with an "Add" button and a list of successors, including "Research tech, regulations, and ...".

11.0 Reflection

The feedback we received was divided into four sections: overall feedback, what we did well, what can be improved, and final thoughts. We thought this feedback structure was very beneficial. The feedback indicated that our report was ‘very well-structured, detailed, and professional.’ We received positive remarks on our structure, realistic ideas, good use of tools, thoughtful risk planning, and the communication plan.

The ‘what can be improved’ section included five subsections. We reviewed each subsection and collectively decided what to implement and what not to.

The first subsection stated we were missing visuals for the Gantt Chart and the Power/Interest grid. Our report had visuals for both, and we suspect that there might have been an error in loading the inserted images properly during the upload or download. Therefore, we will submit the report as a PDF to avoid this issue in the future.

In the second subsection, ‘some sections could be clearer,’ we received feedback that some roles in the stakeholder list, like ‘competitors,’ were unclear. To address this feedback, we added more details to some generalised roles in the stakeholder list. We thought this was helpful feedback, as some roles might be reasonable for us, as we have worked on this project for weeks, but an external stakeholder might not be that familiar with the project.

In the third subsection, ‘formatting tweaks,’ we received feedback that we should clean up our bullet points and change the typo GRANTT Chart to Grantt Chart. Again, we believe that the bullet point feedback originated from an uploading or downloading error, as our version doesn’t show any inconsistency in the bullet points. Uploading the report as a PDF should also fix this problem. We have changed the capitalisation for Grant Chart to address the second point, as we think having a report without typos is important.

The fourth subsection, ‘add more context in places,’ stated that we should briefly explain why Ireland is a good fit for this project in our introduction. We added a short paragraph because we thought the feedback highlighted a gap in our introduction that was important to address. Additionally, the feedback mentioned that we should explain the exclusion from the scope management section more clearly. We added explanatory sentences to every exclusion to help clarify sections of the report.

The last subsection, ‘reflection section is still missing,’ was intended to inform us that we were still missing the reflection section of the report. We are confused about why we received this

feedback, as we had already added a reflection section to the report, noting that it would be written after we received the feedback. Overall, we incorporated most of the received feedback since it highlighted important aspects such as typos and gaps in our coverage.